

Numeric and Algebraic Expressions

mathNexa

Name: _____

Date: _____

Success Criteria:

- Evaluate numerical expressions using order of operations
- Identify parts of an algebraic expression
- Translate verbal phrases into algebraic expressions
- Evaluate algebraic expressions using substitution
- Generate equivalent expressions by simplifying terms

1. What is the value of the numerical expression $4^2 + (15 - 3) \div 3$?

- ☐ A 18 ☐ B 12 ☐ C 28 ☐ D 20

2. What is the value of the numerical expression $3 \times 2^3 - 5 \times 2$?

Answer: _____

3. In the algebraic expression $7x^2 - 4x + 9$, what are the coefficient of the linear term x and the constant term?

- ☐ A Coefficient: 7 , Constant: -4 ☐ B Coefficient: 4 , Constant: 9
☐ C Coefficient: -4 , Constant: 7 ☐ D Coefficient: -4 , Constant: 9

4. A craft store sells art kits containing p paint brushes and 4 jars of paint. What is the simplified algebraic expression for the total number of items in 6 identical art kits?



Answer: _____

5. What is the value of the algebraic expression $\frac{5y + 3}{2}$ when $y = 7$?

A 19

B 16

C 26

D 38

6. A rectangular raised garden bed has a length of $(3x + 2)$ metres and a width of 5 metres. What is the simplified algebraic expression for the perimeter of the garden bed?



Answer: Perimeter = _____ metres

7. Which expression is equivalent to $4(3a + 5) - 2a$ when expanded and simplified?

A $10a + 20$

B $10a + 5$

C $12a + 20$

D $14a + 20$

8. What is the simplified form of the expression $8m + 5n - 3m + 2n - 4$ after combining like terms?

Answer: _____

9. An online sports store charges \$12 per jersey plus a flat shipping fee of \$8. If a team orders j jerseys and receives a \$15 discount voucher off their total order, which expression represents the final amount paid in dollars?



A $20j - 15$

B $12j - 7$

C $12j - 15$

D $12j + 7$

10. The total surface area of an open-top storage bin is given by $A = lw + 2lh + 2wh$. What is the value of A when $l = 6$ cm, $w = 4$ cm, and $h = 5$ cm?

Answer: $A =$ _____ cm^2

Answers

1. D

2. 14

3. D

4. $6p + 24$

5. A

6. $6x + 14$

7. A

8. $5m + 7n - 4$

9. B

10. 124